

Hero Adjusted Plus–Minus in *Marvel Rivals*

Which heroes make a starting draft more likely to win?

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Which heroes make a starting draft more likely to win? Raw win rates mix hero choice with the players selecting it, their teammates, and their opponents. Using 451,204 competitive matches, this report compares predicted outcomes when one starting hero is replaced by another in the same role, accounting for observed rank, map, and draft differences. The comparison follows the match from its opening, including whatever swaps occur afterward.

Three findings organise what follows. First, substantial differences remain between heroes within each role after those adjustments: replacing the average legal same-role alternative with Mantis raises the predicted chance of winning by +6.21 percentage points on average, with The Thing at +5.02 among Tanks and Magik at +4.37 among Damage heroes, while the weakest heroes in each role sit several points below zero. Second, some estimates change considerably across rank environments: Elsa Bloodstone moves from -1.92 points in the lower third of sampled lobbies to $+0.23$ in the upper third, and Black Panther moves the other way, from $+0.03$ to -2.33 . Third, a pair’s extra synergy is a different quantity from the individual value of its members: two heroes can each be good picks and still add less together than their separate values suggest, and the designed team-ups supply the largest positive synergies but not the only ones.

The sample counts are these. 471,008 eligible matches remain after the filters described in Appendix A; 451,204 of them, played up to 4 September 11:00 UTC, were used for the selected fit, and the 19,804 played in the final four days were held back for an internal temporal evaluation. The model is a single penalized logistic regression on the starting drafts; its equation, identification checks, penalty search, and software metadata are in Appendix B. In the main text, what matters is what it adjusts for and what its outputs mean.

What does “starting a hero” mean?

Hero identity is not fixed within a match. Figure 1 shows an illustrative player who starts Spider-Man, plays him for two minutes, then swaps to Elsa Bloodstone for the remaining eight. There are three natural ways to credit that player, and they answer different questions. The starting-hero rule, used throughout this report, credits the whole match to Spider-Man: the question is what happens after a team opens with him. The most-played rule credits Elsa Bloodstone. The time-weighted rule splits the match 20/80.

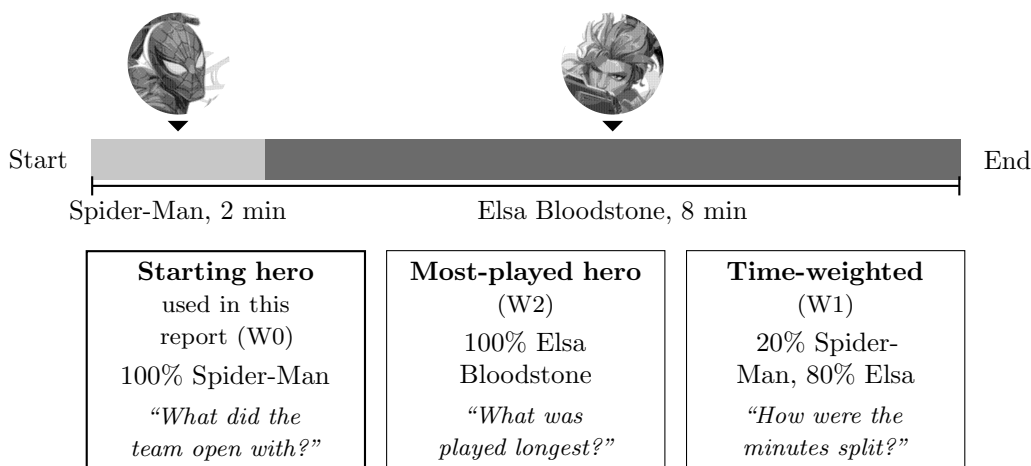


Figure 1: Three ways to credit one player who swaps heroes. Illustrative; no estimation is involved. The starting-hero rule is the one used here. Every draft-derived quantity in the model (hero contrasts, composition, allied pairs, opposing pairs) uses the same starting record.

The distinction matters because switching can respond to how the match develops. A player who is losing a matchup, or who sees the opposing team change composition, may swap; the minutes they end up on each hero are then partly a consequence of the match rather than a description of its starting conditions. The opening and the eventual allocation of minutes therefore answer different questions. This report asks the first one. The starting record carries its own limitation: a player who abandons a hero after thirty seconds still counts as having started it, so the estimates describe the consequences of opening with a hero, swaps included, rather than the hero’s performance in uninterrupted play.

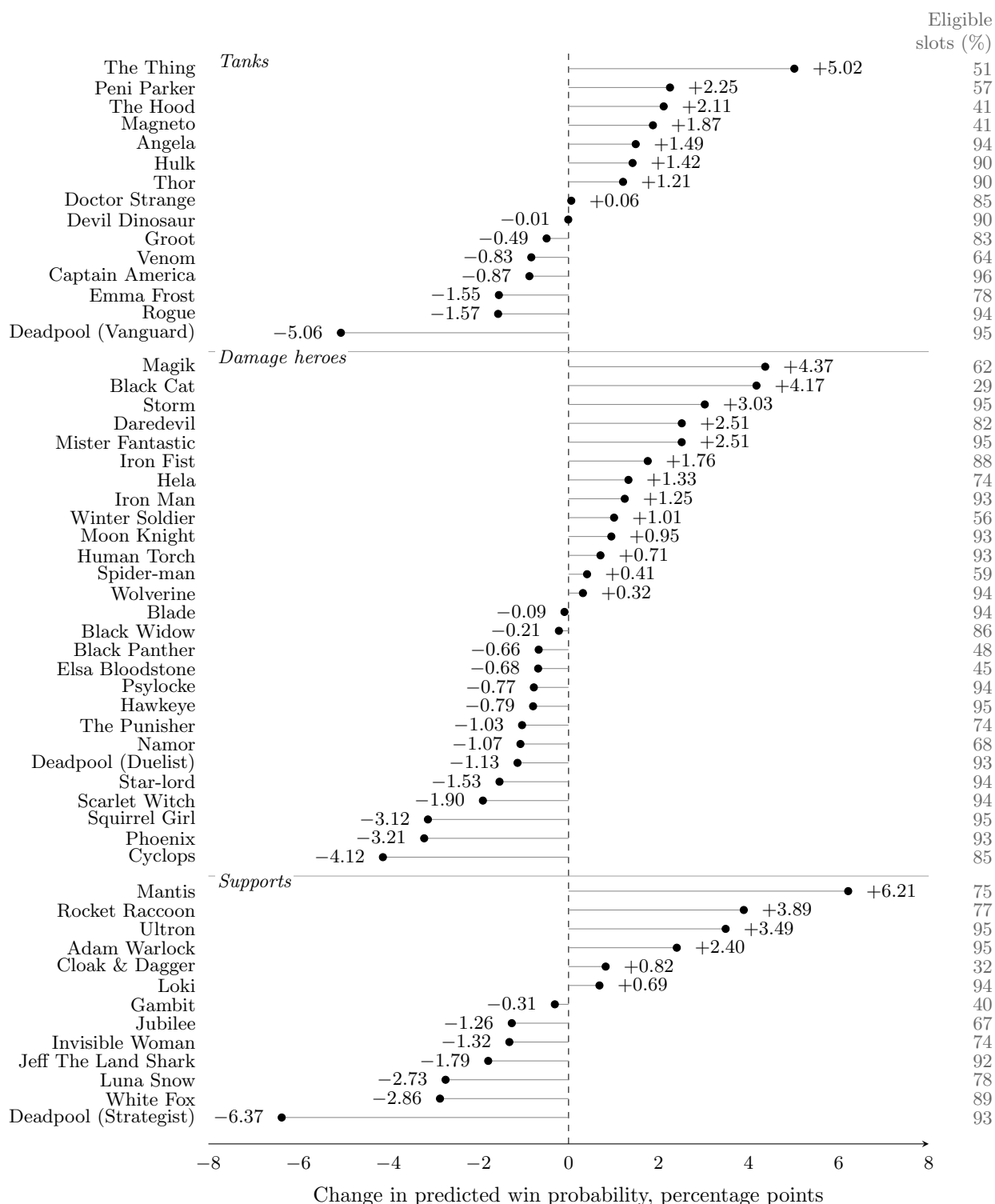
What the model adjusts for

The model predicts the probability that side 0 wins a match from the two starting drafts and pre-match information. Beyond each hero’s own contrast it includes the map, the difference in the two teams’ mean pre-match rank score, the lobby’s overall rank, the role composition of each team (how many Tanks, Damage heroes, and Supports), every pair of allied starters, every pair of opposing starters, and a deviation for each hero on each map. Hero, composition, and pair effects are all allowed to change linearly with lobby rank. With 6,700 coefficients on 451,204 matches, the pair and rank-slope blocks are shrunk toward zero by ridge penalties whose strengths were chosen on chronological development folds. Designated team-ups are labels attached afterward; they receive no special treatment in estimation.

Which heroes have the largest estimated replacement values?

Figure 2 shows the central quantity of the report. For each hero, the model takes actual starting slots occupied by a hero of the same role, puts the hero into that slot instead, keeps the slot’s player, rank, map, side, and the other eleven starters fixed, recomputes every affected term, and averages the change in predicted win probability. The benchmark is the average legal same-role hero for that slot, so a value of +6.21 for Mantis means that, averaged over the slots where Mantis could legally have started, starting Mantis instead of an average available Support raises the predicted win probability by +6.21 percentage points. A positive value is an average model prediction over those contexts, not a bonus that every draft would receive.

Figure 2: Replacement values against the average legal same-role hero, all 55 heroes



Notes. Each dot is the average change in predicted win probability, in percentage points, when the hero replaces the average legal same-role hero in the same starting slot, over 120,000 development matches; heroes are sorted within role. *Eligible slots* is the share of same-role slots in which the hero was a legal replacement (not already on that side and not banned in that match); each hero's average runs over its own eligible subset. No uncertainty has been computed for these values, so close neighbours should not be read as distinguishable.

Within each role the spread is wide. Among Tanks the largest estimates are The Thing (+5.02), Peni Parker (+2.25), and The Hood (+2.11), and the smallest Deadpool (Vanguard) (−5.06), and Rogue (−1.57). Among Damage heroes the largest are Magik (+4.37), Black Cat (+4.17), and Storm (+3.03) and the smallest Cyclops (−4.12), and Phoenix (−3.21). Among Supports, Mantis leads at +6.21, followed by Rocket Raccoon (+3.89), and Ultron (+3.49), with Deadpool (Strategist) (−6.37), and White Fox (−2.86) at the bottom. These are the largest fitted values within their roles; without effect uncertainty the report does not claim that heroes a few tenths of a point apart differ.

Two benchmarks are available, and they can disagree for the same hero. Doctor Strange is the clearest case, shown in Table 1: approximately neutral against the legal same-role reference at +0.06 percentage points, but −1.41 points against the heroes actually selected in the evaluated slots. Both numbers come from the same model. They differ because the alternative being replaced differs: Tank slots in the sample are disproportionately filled by strong Tanks, so replacing the actual incumbent is a harder comparison than replacing an average of everything legal.

Table 1: Two replacement benchmarks for Doctor Strange, from the same fit

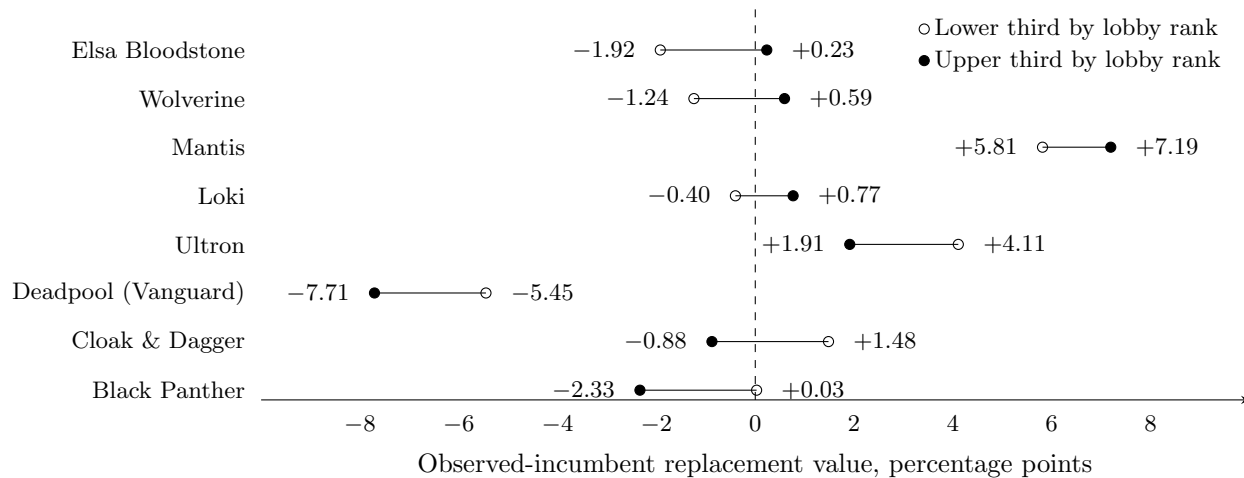
Benchmark	Alternative being replaced	Value, pp
Common reference	the average legal same-role hero in the slot	+0.06
Observed incumbent	the Tank who actually started in the slot	−1.41

The averaging rule deserves a precise statement, because the two benchmarks share a context pool but not a context set. The pool is 120,000 development matches. For a given hero, the contexts are every starting slot in those matches occupied by a same-role hero, and a context is eligible only if the hero was not already on that side and was not banned in that match. Each hero’s value is the mean over its own eligible contexts, so heroes differ in which slots they are scored on. Bans drive most of the difference: Black Cat is eligible in 29% of Damage slots, against 94% for Wolverine, because she is banned so often. The common reference compares the hero with the average of the same-role heroes that were legal in that slot, computed slot by slot, which is why it is the fairer of the two for ranking heroes, while the observed-incumbent benchmark answers the more literal question of what would have happened in the slots as they were actually filled.

How do those values differ across rank environments?

The replacement values can also be averaged within thirds of the sample by lobby rank, the mean pre-match score of the twelve starters, with cutoffs at 4,437 and 4,642. Figure 3 shows the heroes whose observed-incumbent value changes most between the lower and upper thirds: the four largest increases and the four largest decreases, which is the selection rule and the only one. The full roster is in Appendix Figure 7.

Figure 3: Heroes whose replacement value changes most between the lower and upper thirds by lobby rank



Notes. Observed-incumbent replacement values averaged within the lower and upper thirds of sampled lobbies by mean pre-match score (cut at 4,437 and 4,642), for the four largest increases and the four largest decreases among the 55 heroes. Each line joins two estimates from the same fit; it is not a confidence interval, and no uncertainty has been computed for these values. Appendix Figure 7 shows every hero.

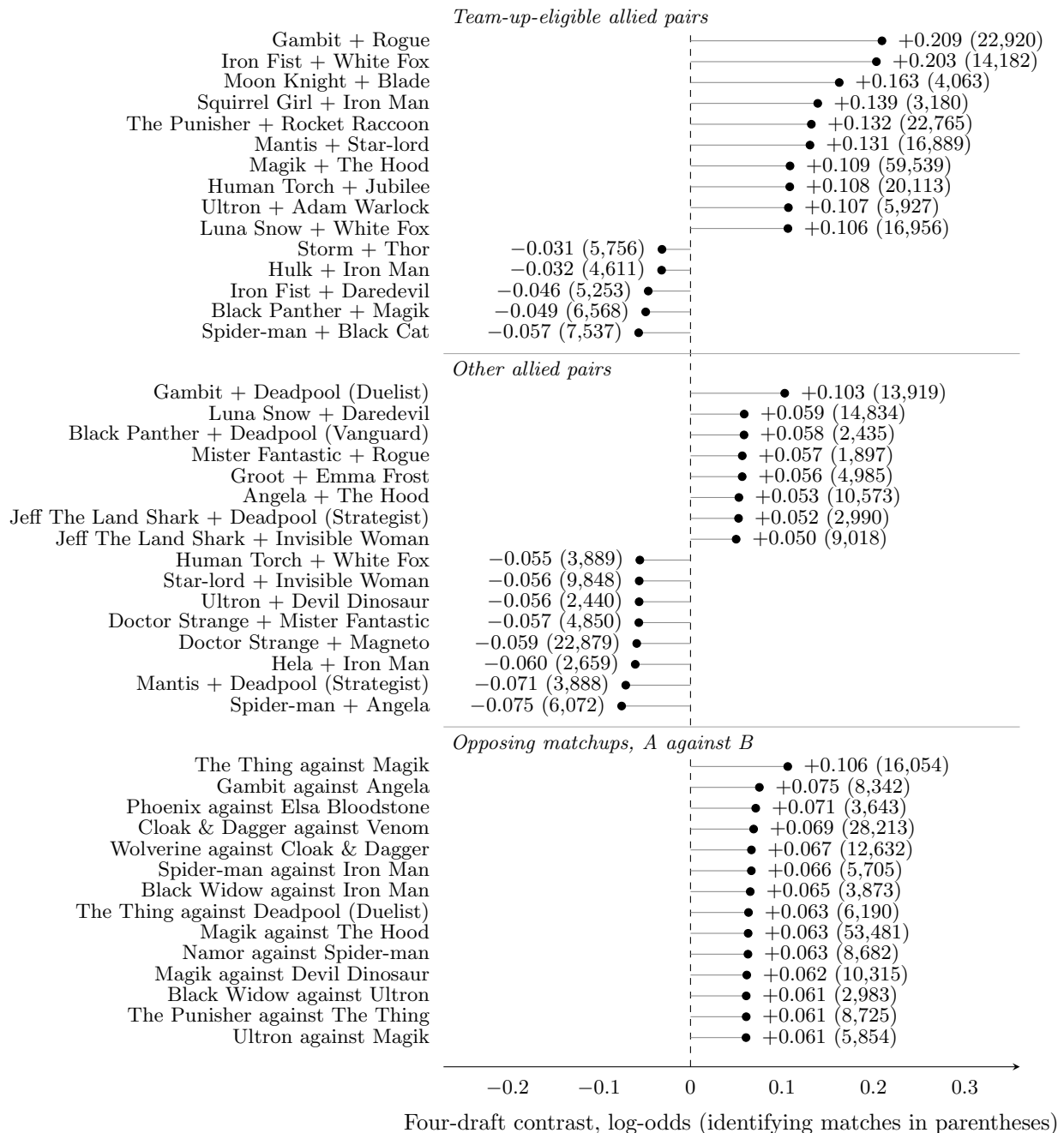
Elsa Bloodstone rises from -1.92 to $+0.23$ points, Wolverine from -1.24 to $+0.59$, and Mantis, already the strongest Support, from $+5.81$ to $+7.19$. In the other direction, Black Panther falls from $+0.03$ to -2.33 , Cloak & Dagger from $+1.48$ to -0.88 , and Ultron from $+4.11$ to $+1.91$. In all, 26 of the 55 heroes move by at least one percentage point between the two thirds.

These are variations across observed rank environments, and they should be read that way. The lobbies in the upper third differ from those in the lower third in more than the players' skill: the compositions, bans, maps, and opponents a hero meets all shift with rank, and the scores average over those contexts as encountered. The figure does not isolate the mechanical effect of raising player skill while holding everything else fixed. It also cannot speak to the top of the ladder, which is thinly sampled: the upper third begins at a lobby mean of 4,642, and few sampled lobbies average above 5,000.

Which teammates fit together, and which opposing matchups stand out?

A pair effect is reported as a four-draft contrast on the log-odds scale. Take the pair's coefficient, subtract the average coefficient of each member with the same-role alternatives to the other, and add back the average over alternative pairs. The result asks whether these two heroes add more together than their separate relationships with other heroes would predict; it cancels the hero main effects and each hero's other pairings, so it does not depend on where additive hero content is placed. These are log-odds contrasts, not percentage-point contributions of the pair to a draft: a four-draft contrast compares four hypothetical drafts, not one draft with and without the pair. Figure 4 shows selected contrasts with the number of identifying matches, meaning distinct development matches in which the pair's signed feature is nonzero; only pairs with at least 500 such matches are shown.

Figure 4: Pair contrasts from the selected fit: designated team-ups, other allied pairs, opposing matchups



Notes. Four-draft contrasts in log-odds, with identifying-match counts in parentheses. Top: allied pairs designated as team-ups in the Season 9.5 catalogue. Middle: other allied pairs. Bottom: opposing matchups, oriented so that “A against B” means A’s side wins more often when the two start on opposite sides than the heroes’ other matchups predict. Selection is by fitted size among pairs with at least 500 identifying matches; no uncertainty has been computed for these contrasts.

The largest fitted team-up-eligible contrasts are Gambit with Rogue (+0.209), Iron Fist with White Fox (+0.203), and Moon Knight with Blade (+0.163). These are the largest among pairs meeting the support rule, not statistically established best combinations, and a positive contrast for a designated pair does not identify the effect of activating the team-up ability, because the data do not record activation. The sign can also run the other way for a designated pair. Black Cat and

Spider-Man both have positive common-reference replacement values (+4.17 and +0.41), while their allied-pair contrast is -0.057 : good individual picks need not provide positive extra synergy together, and this estimate says nothing about their team-up ability in particular.

Among pairs that are not designated team-ups, the largest positive contrasts are Gambit with Deadpool (Duelist) (+0.103), Luna Snow with Daredevil (+0.059), and Black Panther with Deadpool (Vanguard) (+0.058), and the largest negative Spider-man with Angela (-0.075), Mantis with Deadpool (Strategist) (-0.071), and Hela with Iron Man (-0.060). The largest opposing matchups are The Thing against Magik (+0.106), Gambit against Angela (+0.075), Phoenix against Elsa Bloodstone (+0.071), and Cloak & Dagger against Venom (+0.069). The Thing against Magik at +0.106 rests on the most identifying matches of these, 16,054. Any gameplay explanation for a particular sign is tentative: the contrasts measure what happened in matches that opened with those pairs, adaptation included, and the same sign is consistent with complementary kits, with coordinated players choosing the pair, or with the pair being fielded in situations the model does not observe.

How well does the model predict later matches?

Two checks are shown in Figure 5. The left panel uses three chronological development folds: the model is fitted on the first 60, 73, and 87 per cent of development matches by play time and evaluated on the following slice of about 60,000 matches. Across the three folds the unified model averages a validation log loss of 0.66809, against 0.67012 for the published headline specification refitted on the same rows, an improvement of 0.00204. The reduced fits show where that comes from: adding rank slopes and hero-by-map deviations to the hero and composition blocks improves the loss by 0.00072, adding the shrunk allied and opposing pairs improves it by a further 0.00271, and adding pair rank slopes changes it by +0.00001, which is nothing at the displayed precision. The pairs help only because they are shrunk; the same blocks estimated without penalty had made an earlier holdout worse.

The right panel is an internal temporal evaluation rather than a clean future test. The 19,804 matches are those played in the final four days of the snapshot, after 4 September 11:00 UTC. Their outcomes were kept out of every design build, fold, tuning run, and table above, and were first read for the comparison in Figure 5 (Figure 6 describes the same slice), but the same days overlap the final week already examined in the earlier report on the headline specification, so the comparison inherits that prior use. On those matches the unified model’s paired log-loss improvement over the refitted headline is +0.00184, with a player-resampling interval of +0.00151 to +0.00217. By lobby third the improvement is +0.00263 in the lower third, -0.00091 in the middle, and +0.00304 in the upper, the middle third being the one place where the headline specification did slightly better.

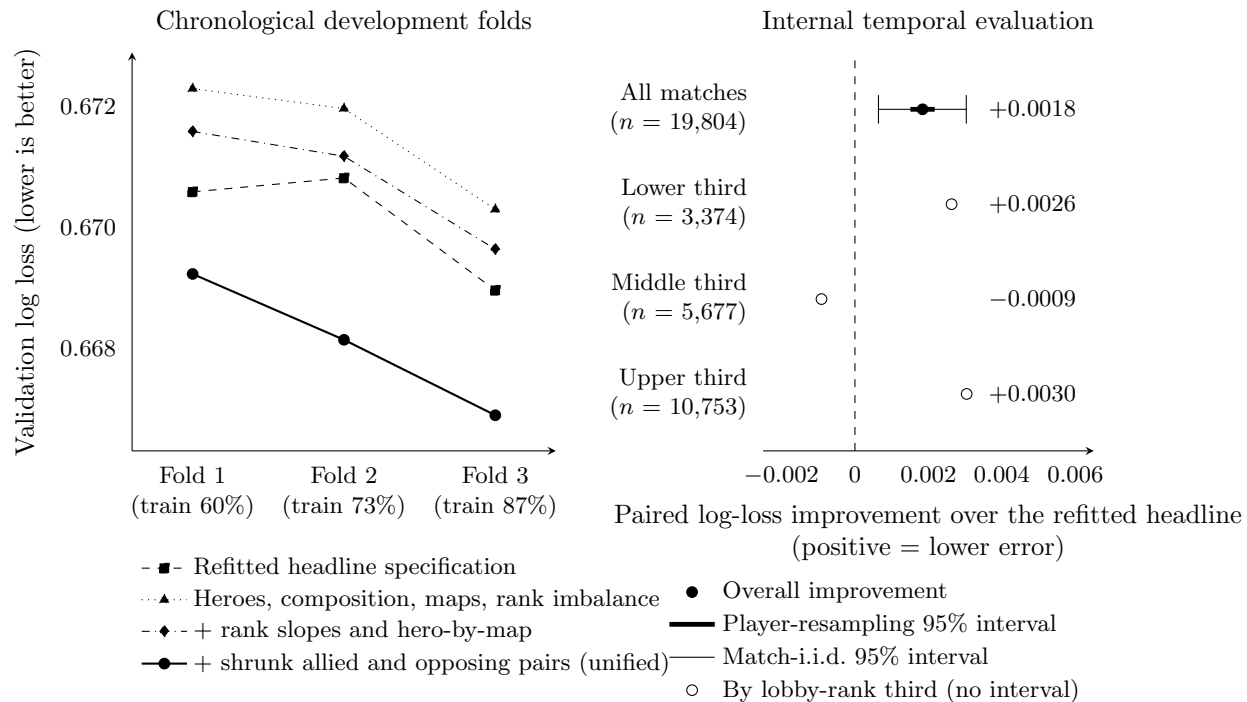
Calibration is summarised in Table 2. A slope of one means predicted probabilities are neither too confident nor too timid; the unified model’s slopes sit near one on the development folds and at 1.062 on the evaluation, rising to 1.116 in the upper third. Saved calibration bins are not available for the unified fit, so no calibration curve is drawn.

Table 2: Calibration slopes of the unified model (ideal value one)

Sample	Overall	Lower third	Middle third	Upper third
Development folds (mean of three)	1.002	0.984	1.022	0.997
Internal temporal evaluation	1.062	0.908	1.038	1.116

Figure 6 looks at the same evaluation slice from the side of the heroes, as the earlier report did for its holdout week. Each of the 19,804 matches contributes two team-sides. Grouping team-sides by each hero they started and comparing the model’s mean predicted win probability with the share that

Figure 5: Does the unified model predict later matches better than the headline specification?



Notes. Left: validation log loss on three chronological development folds for the refitted headline specification and for the unified model with successively larger blocks. Right: paired per-match log-loss improvement of the unified model over the refitted headline on the 19,804 matches of the internal temporal evaluation, overall with two 95% intervals (one treating matches as independent, one resampling players so that matches sharing players move together) and by lobby-rank third without intervals. Positive values mean lower prediction error.

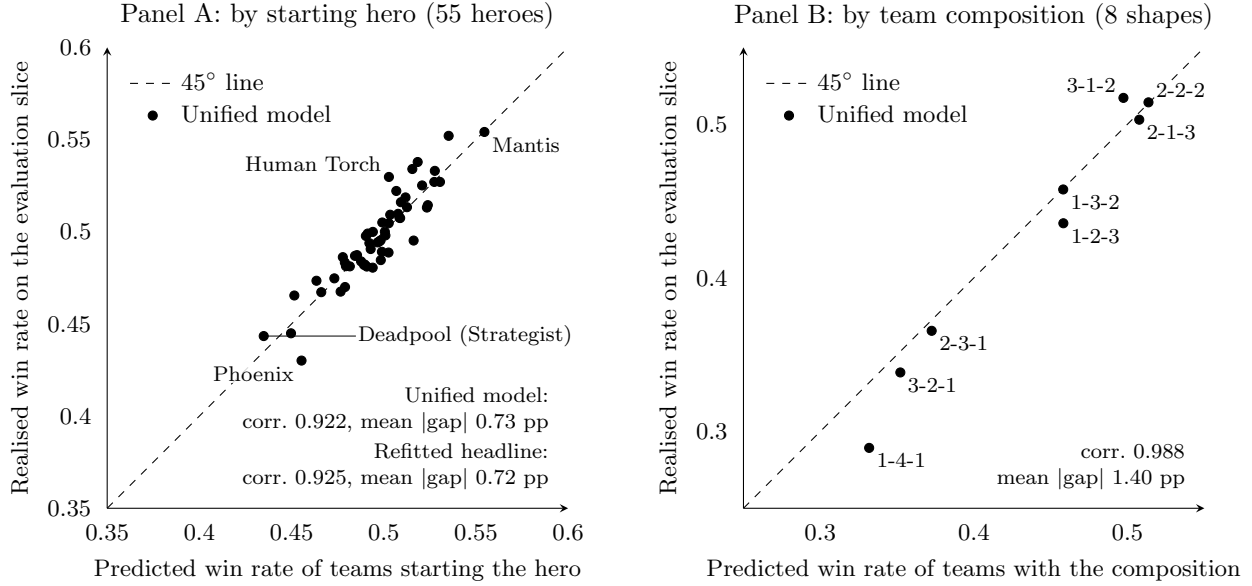
actually won gives one point per hero; grouping by composition shape gives one per shape. Across heroes the correlation is 0.922 and the mean absolute gap 0.73 percentage points, against 0.925 and 0.72 for the refitted headline specification, which shares the hero terms: at this level of aggregation the two models are not distinguishable, and the improvement in Figure 5 is not a matter of ranking heroes better. Across composition shapes the correlation is 0.988.

What remains uncertain?

Two kinds of uncertainty are involved, and only one has been computed. The interval on the evaluation’s average improvement comes from resampling players so that matches sharing players move together. The stability of individual hero and pair estimates under the same resampling is a separate exercise; at the time of this build it had completed 19 replications and is not reported, so no figure in this report carries confidence bands or significance marks, and differences of a few tenths of a point between heroes, or a few hundredths in pair contrasts, should be treated as unresolved.

The estimates describe the sampled competitive environment, not the whole ranked population. Lobby means run from 4,141 at the 5th percentile to 4,944 at the 95th, no sampled player scores above 5,750, and the crawl reaches players only through public profiles, so the top of the ladder is absent and nothing here should be extrapolated to it. Hero choice is also not random: rank score adjusts for broad differences in player strength, not for a player’s skill on the specific hero, and coordinated groups may choose particular pairs. The pair contrasts therefore combine kit complementarity with who plays the pair and when.

Figure 6: Does the model predict which heroes' teams win? By starting hero and by composition



Notes. The 19,804 matches of the internal temporal evaluation, each contributing two team-sides; predictions are from the selected fit on development rows. Panel A groups team-sides by each hero they started: the horizontal axis is the unified model's mean predicted win probability for those team-sides, the vertical axis the share that actually won. Labels mark the two heroes farthest from the line and the two extremes; the smallest hero group has 1,240 team-sides, so a realised share can miss its prediction by a point or two through sampling alone. Panel B groups team-sides by composition shape (Tanks–Damage–Supports) with at least 100 team-sides. The refitted headline specification's statistics are shown for comparison; its points are not drawn because they sit on top of the unified model's.

The sample was checked against the official patch record: the Season 9.5 balance state ran from 7 August to the Season 10 patch on 11 September, and the five updates in between were cosmetic. A weekly win-rate check flags only The Hood, released on 7 August, whose win rate drifted over the season; the audit establishes that no identified balance change coincided with the drift, not what caused it. Matches in which two teammates share the same first-appearance hero, 6,475 in all, were excluded because such a start cannot be simultaneous and the record is unverifiable. Finally, the evaluation above overlaps data used earlier, as disclosed; a clean test needs matches played after the model was fixed, which the Season 10 patch now makes a different balance regime. Refitting the same specification and penalties on all 471,008 eligible matches, including the evaluation slice, changes the common-reference replacement values by 0.03 points on average, with a rank correlation of 0.9997 between the two sets; the tables in this report use the development fit.

Appendix A. Data, regime, and protocol

Snapshot. 477,483 competitive PC ranked matches played between 2026-08-07 11:12 and 2026-09-08 11:00 UTC, frozen on 2026-09-09 with a manifest of file hashes. Filters: twelve players, six per side, no draws, pre-match scores present, every player with positive play time, at least 240 seconds. The starting record is taken from the first-appearance entry of each player's hero list; 6,475 matches with a duplicated first-appearance hero within a side are excluded, leaving 471,008.

Regime. Official patch pages place the snapshot inside one balance regime, from the Season 9.5 patch (7 August 09:00 UTC, which also introduced The Hood) to the Season 10 patch (11 September 09:00 UTC). Updates on 13, 20, and 27 August and 3 September were cosmetic or bug fixes.

Development and evaluation split. Development rows are the 451,204 matches played up to 4 September 11:00 UTC; the evaluation slice is the 19,804 played afterward. Collection was paused before the rebuild, so the slice is internal to the snapshot. The candidate’s specification, penalties, and coefficients were hashed into a lock file (b941d2652002, 2026-09-10 03:35 UTC) before the slice’s outcomes were read, and the evaluator refuses to run twice. The predeclared comparison was a paired log-loss improvement with a player-resampling interval excluding zero, calibration slope between 0.85 and 1.15 overall and by third, and no third worse than -0.001 ; all held. The baseline for the comparison is the headline specification refitted on the development rows, because its published fit had used the whole snapshot. As stated in the main text, the slice overlaps data examined in the earlier report.

Rank coverage. Table 3 gives the distribution of lobby means and individual scores. 14,722 matches have a lobby mean above 5,000 and 123 distinct players score above 5,500; no player scores above 5,750.

Table 3: Rank coverage of the development snapshot (pre-match scores)

	p5	p25	p50	p75	p95	p99	max
Lobby mean score	4,141	4,379	4,548	4,707	4,944	5,126	5,525
Highest player in lobby	4,216	4,463	4,627	4,784	5,023	5,217	5,657
Individual player score	4,133	4,376	4,547	4,713	4,950	5,135	5,657

Appendix B. Model, identification, and penalty search

Let A_{mh} and B_{mh} indicate whether hero h starts on side 0 or side 1 of match m , r_m the standardized lobby rank, ΔR_m the standardized difference in team mean pre-match score, v_m the map, and $c(\cdot)$ the composition. With $x_{mh} = A_{mh} - B_{mh}$, $z_{mhh} = A_{mh}A_{mh} - B_{mh}B_{mh}$, and $w_{mhh} = A_{mh}B_{mh} - A_{mh}B_{mh}$ for $h < k$,

$$\begin{aligned} \text{logit } p_m = & \alpha_{v_m} + (\theta_0 + \theta_1 r_m) \Delta R_m + (\gamma_0 + \gamma_1 r_m)^\top [c(A_m) - c(B_m)] + \sum_h (\beta_h + d_h r_m + u_{h,v_m}) x_{mh} \\ & + \sum_{h < k} (S_{hk} + T_{hk} r_m) z_{mhh} + \sum_{h < k} (C_{hk} + D_{hk} r_m) w_{mhh}. \end{aligned}$$

The allied block is symmetric and the opposing block antisymmetric, so swapping the two camps negates every feature except the map intercept; the implementation checks this numerically. Every observed composition shape, allied pair, opposing pair, and hero-by-map cell has a coefficient. The blocks are aliased in known ways: summing a hero’s allied features over five teammates gives five times its own contrast, summing its opposing features over six opponents gives six times it, summing allied features within a role pair gives a function of the composition contrast, and summing hero-by-map deviations over maps gives the hero contrast again. Each candidate centering direction was tested against the data and imposed only when its image lies exactly in the span of the lower-order blocks; 187 of 187 candidates did, the largest residual being 7.4×10^{-15} , leaving 6,700 free coefficients. Five ridge groups shrink the coefficients on their native scale: hero and composition averages (λ_1), allied averages (λ_2), opposing averages (λ_3), hero and composition rank slopes with hero-by-map deviations (λ_4), and pair rank slopes (λ_5); map intercepts and the rank-imbalance terms are unpenalized. The rank basis and ΔR are scaled with training rows only.

Penalties were tuned on the three chronological folds by a bounded coordinate search on $\log_{10} \lambda$ with warm starts, 39 evaluations in all, the coarse passes on a half subsample of each fold’s training rows and the fine pass on full rows. The selected penalties are $\lambda = (10, 900, 1000, 1000, 30000)$. The

surface near the optimum is flat: the eight best full-row evaluations in Table 4 lie within 0.00006 of each other. The selected fit converged to a coefficient change below 10^{-6} in 4 Newton steps. Table 5 gives the reduced-model ladder behind Figure 5.

Table 4: Best full-row evaluations of the penalty search

$\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5$	Mean validation log loss	Mean calibration slope	Converged
10,900,1000,1000,30000	0.66809	1.002	yes
3.33333,900,1000,1000,30000	0.66809	1.000	yes
30,900,1000,1000,30000	0.66809	1.004	yes
10,900,1000,1000,90000	0.66809	1.002	yes
10,900,1000,1000,1e+09	0.66809	1.002	yes
10,900,1000,1000,10000	0.66811	0.999	yes
10,900,1000,3000,30000	0.66814	1.010	yes
10,900,1000,3000,90000	0.66814	1.010	yes

Table 5: Development ladder: genuine reduced fits at the selected penalties

Model	Free parameters	Fold 1	Fold 2	Fold 3	Mean	Cal. slope
Headline spec., refitted on development rows	187	0.67059	0.67082	0.66896	0.67012	—
Heroes, composition, maps, rank imbalance	96	0.67230	0.67197	0.67030	0.67152	1.009
+ rank slopes and hero-by-map	984	0.67159	0.67118	0.66964	0.67080	1.001
+ shrunk allied and opposing pairs	3,842	0.66924	0.66814	0.66690	0.66809	1.002
+ pair rank slopes (selected fit)	6,700	0.66923	0.66814	0.66689	0.66809	1.002

Software metadata. Design and fitter in the repository’s `lineup` package at revision 832c615; penalty search run on an 8-CPU Colab runtime from the cached design; report build `d8aa3352`, the hash of this document’s text with the hash itself left blank.

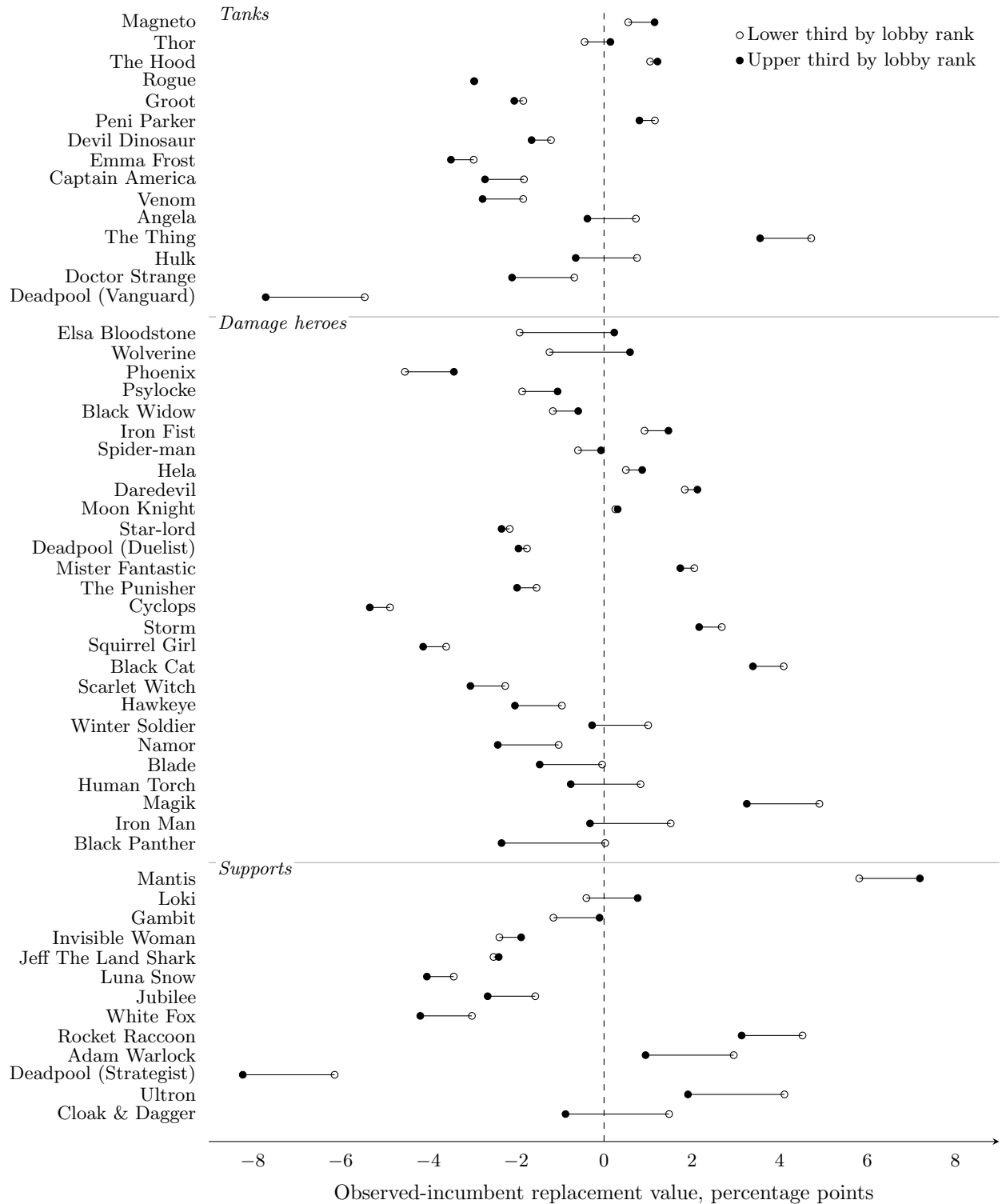
Appendix C. Full tables and figures

Table 6: Hero coefficients and replacement values, all 55 heroes

Hero	Coefficient		Replacement, pp		Obs. incumbent			Hero	Coefficient		Replacement, pp		Obs. incumbent		
	β_h	slope	Common ref.	Obs. incumbent	Lower third	Upper third	Slots %		β_h	slope	Common ref.	Obs. incumbent	Lower third	Upper third	Slots %
<i>Panel A: Tank (15 heroes)</i>								<i>Panel B: Damage (27 heroes)</i>							
The Thing	+0.177	-0.011	+5.02	+4.13	+4.72	+3.56	51	Magik	+0.103	-0.017	+4.37	+4.07	+4.91	+3.25	62
Peni Parker	+0.096	+0.001	+2.25	+0.95	+1.16	+0.81	57	Black Cat	+0.128	-0.010	+4.17	+3.88	+4.09	+3.39	29
The Hood	+0.052	+0.021	+2.11	+1.15	+1.05	+1.22	41	Storm	+0.131	-0.002	+3.03	+2.42	+2.68	+2.17	95
Magneto	+0.081	+0.026	+1.87	+0.81	+0.55	+1.15	41	Daredevil	+0.039	+0.004	+2.51	+1.95	+1.84	+2.13	82
Angela	+0.042	-0.009	+1.49	+0.14	+0.73	-0.38	94	Mister Fantastic	+0.088	-0.007	+2.51	+1.88	+2.06	+1.74	95
Hulk	+0.073	-0.007	+1.42	+0.06	+0.75	-0.65	90	Iron Fist	+0.055	+0.012	+1.76	+1.17	+0.92	+1.47	88
Thor	-0.049	+0.016	+1.21	-0.15	-0.44	+0.14	90	Hela	+0.087	+0.020	+1.33	+0.70	+0.50	+0.87	74
Doctor Strange	-0.025	-0.009	+0.06	-1.41	-0.68	-2.10	85	Iron Man	+0.083	-0.020	+1.25	+0.60	+1.52	-0.32	93
Devil Dinosaur	-0.013	+0.006	-0.01	-1.45	-1.21	-1.65	90	Winter Soldier	+0.045	-0.015	+1.01	+0.42	+1.01	-0.27	56
Groot	-0.032	+0.006	-0.49	-1.96	-1.84	-2.04	83	Moon Knight	+0.040	+0.012	+0.95	+0.29	+0.26	+0.31	93
Venom	-0.040	-0.009	-0.83	-2.34	-1.84	-2.77	64	Human Torch	-0.004	-0.020	+0.71	+0.05	+0.83	-0.76	93
Captain America	-0.070	-0.010	-0.87	-2.28	-1.82	-2.71	96	Spider-man	-0.003	+0.012	+0.41	-0.32	-0.59	-0.07	59
Emma Frost	-0.098	-0.000	-1.55	-3.25	-2.97	-3.49	78	Wolverine	-0.028	+0.036	+0.32	-0.35	-1.24	+0.59	94
Rogue	-0.075	+0.008	-1.57	-2.97	-2.95	-2.96	94	Blade	-0.028	-0.021	-0.09	-0.78	-0.04	-1.46	94
Deadpool (Vanguard)	-0.218	-0.031	-5.06	-6.60	-5.45	-7.71	95	Black Widow	-0.001	+0.012	-0.21	-0.91	-1.17	-0.59	86
<i>Panel C: Support (13 heroes)</i>								Black Panther	-0.009	-0.036	-0.66	-1.47	+0.03	-2.33	48
Mantis	+0.267	+0.032	+6.21	+6.44	+5.81	+7.19	75	Elsa Bloodstone	-0.002	+0.040	-0.68	-1.24	-1.92	+0.23	45
Rocket Raccoon	+0.175	-0.004	+3.89	+3.87	+4.52	+3.13	77	Psylocke	-0.018	+0.017	-0.77	-1.48	-1.86	-1.06	94
Ultron	+0.151	-0.023	+3.49	+3.06	+4.11	+1.91	95	Hawkeye	-0.052	-0.013	-0.79	-1.49	-0.96	-2.03	95
Adam Warlock	+0.101	-0.033	+2.40	+1.96	+2.95	+0.95	95	The Punisher	-0.042	-0.005	-1.03	-1.77	-1.54	-1.98	74
Cloak & Dagger	+0.010	-0.019	+0.82	+0.29	+1.48	-0.88	32	Namor	-0.028	-0.015	-1.07	-1.81	-1.03	-2.42	68
Loki	+0.022	+0.035	+0.69	+0.21	-0.40	+0.77	94	Deadpool (Duelist)	-0.062	+0.001	-1.13	-1.86	-1.76	-1.95	93
Gambit	+0.030	+0.019	-0.31	-0.71	-1.15	-0.10	40	Star-lord	-0.041	-0.006	-1.53	-2.25	-2.15	-2.33	94
Jubilee	-0.066	-0.007	-1.26	-2.09	-1.57	-2.65	67	Scarlet Witch	-0.078	-0.002	-1.90	-2.64	-2.25	-3.04	94
Invisible Woman	-0.070	+0.022	-1.32	-2.12	-2.38	-1.89	74	Squirrel Girl	-0.146	+0.001	-3.12	-3.86	-3.60	-4.12	95
Jeff The Land Shark	-0.099	+0.018	-1.79	-2.42	-2.52	-2.40	92	Phoenix	-0.129	+0.025	-3.21	-3.98	-4.54	-3.42	93
Luna Snow	-0.119	+0.003	-2.73	-3.70	-3.42	-4.03	78	Cyclops	-0.128	-0.004	-4.12	-5.11	-4.87	-5.33	85
White Fox	-0.123	-0.016	-2.86	-3.60	-3.01	-4.19	89								
Deadpool (Strategist)	-0.279	-0.026	-6.37	-7.17	-6.14	-8.23	93								

Notes. All 55 heroes are listed, sorted within role by the common-reference value. β_h is the within-role log-odds contrast at the median lobby score of 4,542, and *slope* its change per standard deviation of lobby rank (252 points). *Common ref.* and *Obs. incumbent* are the two replacement benchmarks of the main text, in percentage points of win probability over 120,000 development matches; *Lower third* and *Upper third* average the observed-incumbent value within the lower and upper thirds of the sample by lobby rank (cut at 4,437 and 4,642), as in Figure 7. *Slots %* is the share of same-role slots in which the hero was a legal replacement; every hero was scored on at least 145,140 eligible slots. No uncertainty has been computed for these values.

Figure 7: Replacement values by lobby rank: lower third against upper third, all 55 heroes



Notes. Observed-incumbent replacement values from the selected fit, averaged within the lower and upper thirds of sampled lobbies by mean pre-match score (cut at 4,437 and 4,642). Within each role, heroes are sorted by the change from the lower to the upper third. Lines join two estimates from the same fit and are not confidence intervals; no uncertainty has been computed for these values.

Table 7: Team-up-eligible allied pairs with at least 500 identifying matches, all 100 pairs

Pair	Contrast	Matches	Pair	Contrast	Matches	Pair	Contrast	Matches
Gambit + Rogue	+0.209	22,920	Rocket Raccoon + Mister Fantastic	+0.039	10,499	Black Panther + Peni Parker	+0.010	2,557
Iron Fist + White Fox	+0.203	14,182	Venom + Jeff The Land Shark	+0.039	8,644	Loki + Hela	+0.010	5,953
Moon Knight + Blade	+0.163	4,063	Invisible Woman + The Thing	+0.039	29,386	Hela + Namor	+0.009	10,547
Squirrel Girl + Iron Man	+0.139	3,180	Hela + Phoenix	+0.037	4,617	Captain America + Thor	+0.007	2,437
The Punisher + Rocket Raccoon	+0.132	22,765	Doctor Strange + Magik	+0.037	24,603	Mister Fantastic + The Thing	+0.006	5,027
Mantis + Star-lord	+0.131	16,889	Mantis + Adam Warlock	+0.035	4,275	Cloak & Dagger + Psylocke	+0.003	15,160
Magik + The Hood	+0.109	59,539	Squirrel Girl + Spider-man	+0.035	3,125	Groot + Jeff The Land Shark	+0.002	4,413
Human Torch + Jubilee	+0.108	20,113	The Punisher + Daredevil	+0.034	3,862	The Punisher + Winter Soldier	+0.000	10,666
Ultron + Adam Warlock	+0.107	5,927	Hawkeye + Cloak & Dagger	+0.033	13,883	Rocket Raccoon + Squirrel Girl	-0.002	6,010
Luna Snow + White Fox	+0.106	16,956	Groot + Star-lord	+0.033	5,188	The Punisher + Devil Dinosaur	-0.004	6,134
Hulk + Doctor Strange	+0.096	8,175	Doctor Strange + Scarlet Witch	+0.033	8,960	Human Torch + The Thing	-0.004	11,499
Storm + Jeff The Land Shark	+0.096	7,545	Black Widow + Phoenix	+0.032	4,041	Venom + Spider-man	-0.004	19,990
Cloak & Dagger + The Hood	+0.092	94,581	Storm + Black Panther	+0.031	2,787	Black Panther + Black Cat	-0.005	2,662
Doctor Strange + Invisible Woman	+0.092	40,971	Black Widow + Daredevil	+0.030	5,429	Moon Knight + Elsa Bloodstone	-0.006	3,089
Loki + Mantis	+0.090	6,996	Ultron + Peni Parker	+0.029	2,422	Mister Fantastic + Invisible Woman	-0.007	9,144
Jubilee + The Hood	+0.086	114,785	Star-lord + Adam Warlock	+0.029	2,006	Psylocke + Emma Frost	-0.010	8,548
Rocket Raccoon + Peni Parker	+0.083	13,302	Loki + Angela	+0.028	3,104	Venom + Blade	-0.010	5,306
Mantis + Groot	+0.071	13,159	Captain America + Winter Soldier	+0.027	7,579	Magneto + Emma Frost	-0.013	36,898
Phoenix + Cyclops	+0.068	8,596	The Thing + Iron Fist	+0.025	14,162	Rocket Raccoon + Groot	-0.016	13,270
Scarlet Witch + Jubilee	+0.067	16,017	Human Torch + Invisible Woman	+0.024	14,581	Thor + Angela	-0.017	2,685
Winter Soldier + Elsa Bloodstone	+0.063	16,804	Hulk + Captain America	+0.023	2,810	Hulk + Wolverine	-0.017	4,536
Storm + Adam Warlock	+0.052	2,124	Jeff The Land Shark + Devil Dinosaur	+0.023	6,651	Magneto + Scarlet Witch	-0.018	18,147
Mantis + Emma Frost	+0.052	28,531	Wolverine + Cyclops	+0.021	3,675	Blade + Jubilee	-0.019	14,149
Spider-man + Peni Parker	+0.050	5,978	Captain America + Blade	+0.021	3,137	Luna Snow + Adam Warlock	-0.029	5,084
Cloak & Dagger + Moon Knight	+0.046	19,222	Magneto + Gambit	+0.020	62,728	Hawkeye + Black Widow	-0.029	2,174
White Fox + Black Cat	+0.046	9,483	Ultron + Iron Man	+0.018	4,055	Storm + Human Torch	-0.030	2,461
Venom + Phoenix	+0.045	10,303	Luna Snow + Emma Frost	+0.016	34,932	Gambit + Cyclops	-0.030	20,331
Elsa Bloodstone + Devil Dinosaur	+0.045	9,421	Star-lord + Angela	+0.015	2,782	Storm + Thor	-0.031	5,756
Iron Man + Thor	+0.044	6,839	Hulk + Namor	+0.015	9,394	Hulk + Iron Man	-0.032	4,611
Magneto + Rogue	+0.044	15,160	Gambit + Jubilee	+0.015	32,597	Iron Fist + Daredevil	-0.046	5,253
Hela + Thor	+0.043	10,438	Cloak & Dagger + Luna Snow	+0.014	35,050	Black Panther + Magik	-0.049	6,568
Wolverine + Gambit	+0.041	10,108	Phoenix + Rogue	+0.013	2,444	Spider-man + Black Cat	-0.057	7,537
Hela + Venom	+0.041	9,967	Psylocke + White Fox	+0.013	4,989			
Hawkeye + Psylocke	+0.039	3,584	Luna Snow + Namor	+0.012	14,262			

Notes. Pairs are listed in descending order of contrast, reading down each column in turn. *Contrast* is the four-draft log-odds contrast at the median lobby rank and *Matches* the number of distinct development matches with a nonzero signed feature. Read as the synergy of a team-up-eligible starting pair; activation of the team-up ability is not observed.

Table 8: Other allied pairs: 25 largest positive and 25 largest negative contrasts

<i>Largest positive</i>				<i>Largest negative</i>			
Pair	Contrast	Matches	Top third %	Pair	Contrast	Matches	Top third %
Gambit + Deadpool (Duelist)	+0.103	13,919	37	Spider-man + Angela	−0.075	6,072	38
Luna Snow + Daredevil	+0.059	14,834	46	Mantis + Deadpool (Strategist)	−0.071	3,888	36
Black Panther + Deadpool (Vanguard)	+0.058	2,435	30	Hela + Iron Man	−0.060	2,659	29
Mister Fantastic + Rogue	+0.057	1,897	28	Doctor Strange + Magneto	−0.059	22,879	37
Groot + Emma Frost	+0.056	4,985	31	Doctor Strange + Mister Fantastic	−0.057	4,850	27
Angela + The Hood	+0.053	10,573	35	Ultron + Devil Dinosaur	−0.056	2,440	22
Jeff The Land Shark + Deadpool (Strategist)	+0.052	2,990	23	Star-lord + Invisible Woman	−0.056	9,848	34
Jeff The Land Shark + Invisible Woman	+0.050	9,018	30	Human Torch + White Fox	−0.055	3,889	26
The Punisher + Invisible Woman	+0.049	15,011	33	Emma Frost + Gambit	−0.053	26,315	32
Hulk + Deadpool (Strategist)	+0.049	5,149	33	Jubilee + Rogue	−0.051	11,643	25
Storm + Devil Dinosaur	+0.046	2,855	25	Adam Warlock + Deadpool (Strategist)	−0.051	1,187	30
Loki + Psylocke	+0.046	2,088	44	Loki + Adam Warlock	−0.050	818	46
Loki + Gambit	+0.046	6,326	45	Hela + Ultron	−0.049	3,179	37
Luna Snow + Psylocke	+0.046	8,908	40	Magik + The Thing	−0.049	14,601	38
Magneto + Wolverine	+0.046	16,482	48	Namor + Gambit	−0.049	13,436	29
Loki + Cloak & Dagger	+0.046	8,641	37	Wolverine + The Thing	−0.048	4,489	45
Magik + Star-lord	+0.045	4,706	35	Peni Parker + Gambit	−0.047	6,877	32
Venom + Elsa Bloodstone	+0.045	6,936	32	Mantis + Devil Dinosaur	−0.047	7,705	34
Jeff The Land Shark + Deadpool (Duelist)	+0.044	4,462	28	Human Torch + Jeff The Land Shark	−0.047	3,374	32
Mantis + Elsa Bloodstone	+0.043	10,927	39	Daredevil + Rogue	−0.045	3,897	40
Angela + Devil Dinosaur	+0.042	1,461	28	Loki + Ultron	−0.045	651	33
Captain America + Hela	+0.042	2,843	37	Iron Fist + Jubilee	−0.045	21,512	33
Namor + Invisible Woman	+0.042	15,997	28	The Punisher + Loki	−0.044	3,226	38
Devil Dinosaur + Deadpool (Strategist)	+0.042	4,645	20	Adam Warlock + White Fox	−0.043	1,735	23
Magik + Deadpool (Strategist)	+0.041	9,146	29	Squirrel Girl + Cyclops	−0.043	2,385	11

Notes. *Contrast* and *Matches* as in Table 7; *Top third %* is the share of the identifying matches in the upper lobby-rank third. Pairs with at least 500 identifying matches that are not designated team-ups.

Table 9: Opposing matchups: 50 largest contrasts, oriented as A against B

Matchup	Contrast	Matches	Top third %	Matchup	Contrast	Matches	Top third %
The Thing against Magik	+0.106	16,054	38	Black Panther against Mantis	+0.053	9,327	46
Gambit against Angela	+0.075	8,342	36	Mister Fantastic against Black Widow	+0.053	2,976	35
Phoenix against Elsa Bloodstone	+0.071	3,643	24	Rocket Raccoon against Squirrel Girl	+0.052	6,015	17
Cloak & Dagger against Venom	+0.069	28,213	31	Namor against Gambit	+0.052	13,764	29
Wolverine against Cloak & Dagger	+0.067	12,632	38	Gambit against Daredevil	+0.051	13,091	41
Spider-man against Iron Man	+0.066	5,705	26	Mister Fantastic against Black Cat	+0.051	2,967	21
Black Widow against Iron Man	+0.065	3,873	29	Black Widow against Hela	+0.051	9,464	48
The Thing against Deadpool (Duelist)	+0.063	6,190	36	Captain America against Invisible Woman	+0.050	7,080	31
Magik against The Hood	+0.063	53,481	34	Peni Parker against Magik	+0.050	5,887	32
Namor against Spider-man	+0.063	8,682	32	The Thing against Rogue	+0.050	5,708	34
Magik against Devil Dinosaur	+0.062	10,315	27	Iron Fist against Black Widow	+0.050	7,097	46
Black Widow against Ultron	+0.061	2,983	38	Mister Fantastic against Magneto	+0.050	14,966	34
The Punisher against The Thing	+0.061	8,725	34	Winter Soldier against Hulk	+0.048	17,460	43
Ultron against Magik	+0.061	5,854	32	Spider-man against Rocket Raccoon	+0.048	26,191	37
Deadpool (Duelist) against Invisible Woman	+0.060	12,725	35	Spider-man against Human Torch	+0.048	7,436	40
Iron Fist against Rocket Raccoon	+0.060	16,582	40	Hulk against Wolverine	+0.048	3,531	47
Storm against Iron Fist	+0.060	3,805	38	The Thing against Mister Fantastic	+0.048	4,781	31
Black Cat against Mantis	+0.058	7,919	28	Psylocke against Moon Knight	+0.048	2,397	23
Venom against Squirrel Girl	+0.056	3,247	15	Iron Fist against Invisible Woman	+0.048	18,123	40
Daredevil against Black Cat	+0.056	4,371	30	Daredevil against Rocket Raccoon	+0.047	14,532	45
Hawkeye against Storm	+0.055	1,684	29	The Hood against The Punisher	+0.047	28,208	31
Daredevil against Invisible Woman	+0.054	16,320	44	Emma Frost against Black Widow	+0.047	13,188	38
Invisible Woman against Winter Soldier	+0.054	47,155	40	Devil Dinosaur against Gambit	+0.047	12,794	27
Captain America against Spider-man	+0.054	4,608	32	Jubilee against Iron Fist	+0.047	23,217	32
Magneto against Hela	+0.053	40,218	46	Gambit against Jeff The Land Shark	+0.047	12,936	27

Notes. Columns as in Table 8. “A against B” means that A’s side wins more often when the two start on opposite sides than the heroes’ other matchups predict.

Appendix D. Earlier-specification results on attribution rules

The numbers here come from the earlier report’s headline specification (starting-hero attribution, within-role constraint, map intercepts, designated team-ups only), not from the unified model, and are kept for reference. Fitting that specification under the three attribution rules of Figure 1 on the same 477,483 matches gave cross-hero standard deviations of the within-role estimates of 2.60 percentage points under starting-hero attribution, 4.27 under most-played attribution, and 6.38 under time weighting; Magik’s estimate moved from +3.48 to +7.94 to +14.57. The rules that use more information from after the opening produce larger estimates, which is consistent with swaps responding to the match, and is the reason the starting record is used here.